RANDOM VARIABLES – Lectures 5–6

$X : S \longrightarrow \mathbb{R}$

A real-valued function X, defined on the sample space S is called a random variable (r.v.)

Experiment $\longmapsto$ Event

The introduction of a r.v. related to an experiment facilitates describing the event of interest.

Discrete random variable

$X : S \longrightarrow \mathbb{N}$ (or any countable set).

$S = \{(H,T), (T,T), (H,H), (T,H)\}$

$X =$ The number of heads.

$X : S \longrightarrow \{0, 1, 2\}$

$P(X=0) = P(\{T,T\}) = \frac{1}{4}$

$P(X=1) = P(\{(H,T), (T,H)\}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

$P(X=2) = P((H,H)) = \frac{1}{4}$

Discrete r.v.

$$X \begin{pmatrix} x_1 & x_2 & \dots & x_n & \dots \\ p_1 & p_2 & \dots & p_n & \dots \end{pmatrix}$$

$p_i \overset{\text{Def.}}{\equiv} P(\{X = x_i\}) \overset{\text{Definition}}{=\!=} f(x_i)$

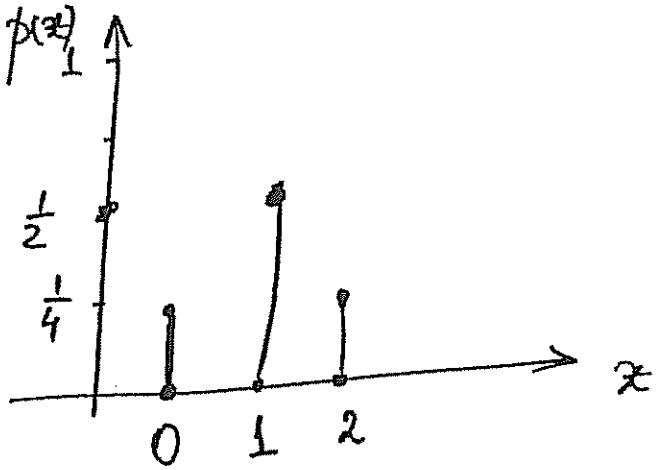
$p_i \geq 0$ for all $i = 1, 2, \dots$

$\sum_{i=1}^{\infty} p_i = 1$.

- $S = \{(H,H), (H,T), (T,H), (T,T)\}$.

X = # of heads.

$$X \begin{pmatrix} 0 & 1 & 2 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$$



- $f(x)$ = probability "mass" function = prob. function = p.f.

" X has probability mass at 0, 1, 2. "

Theorem If X has a discrete distribution,

$$P(X \in C) = \sum_{x_i \in C} f(x_i), \text{ for}$$

any subset C of the real line.

IMPORTANT $P(X \in C) \overset{\text{Notation}}{=\!=} P(s \in S : X(s) \in C)$

① **The Bernoulli random variable**

$$Z \begin{pmatrix} 0 & 1 \\ 1-p & p \end{pmatrix}; \quad P(Z=1) = p$$

- Z has the Bernoulli distribution with parameter p.

② **Uniform distributions on integers**

Let $a \leq b$ be integers. Suppose that the value of a random variable X is equally likely to be any of the integers $a, \ldots, b$. We then say that X has the uniform distribution on integers $a, \ldots, b$.

. There are $b-a+1$ integers from a to b.
If X has the unif. dists. on integers,
then the p.f. of X is:

$$f(x) = \begin{cases} \frac{1}{b-a+1}, & \text{for } x = a, \dots, b \\ 0, & \text{otherwise.} \end{cases}$$

Example → How to associate a random variable to an experiment.

• "Experiment" = Play the lottery game in which you select a 3-digit number (leading 0s allowed).
⟺ Select, independently, 3 balls from 3 urns, each urn containing 10 balls.

• Sample space $= \{(i_1, i_2, i_3) : i_1 \in \{0, \dots, 9\}, i_2 \in \{0, \dots, 9\}, i_3 \in \{0, \dots, 9\}\}$

$= S$

• A winning ticket = An ordered triple.

- $|S| = 10^3 = 1000$.
- We can associate to each $S = (i_1, i_2, i_3)$ a number $x \in \{0, 1, 2, \ldots, 999\}$ via the following function:

$$X(s) = 100 i_1 + 10 i_2 + i_3.$$

- Therefore, calculating the probability that a particular triple is selected (order matters!) is equivalent to calculating the probability that the value of X is one of the numbers $\{0, \ldots, 999\}$. The p.f. is below

$$X \begin{pmatrix} 0 & 1 & 2 & 3 & \cdots & 999 \\ \frac{1}{1000} & \frac{1}{1000} & \frac{1}{1000} & \frac{1}{1000} & \cdots & \frac{1}{1000} \end{pmatrix}.$$

- X is uniform on integers $\{0, \ldots, 999\}$.

VERY IMPORTANT

- **Random variables can have the same distribution without being the same.**

Example. A school has 1000 students, numbered 0, ..., 999 (or 1, ..., 1000). Select one at random. Let Y be the random variable that records the selected number: $Y \begin{pmatrix} 0 & 1 & \cdots & 999 \\ \frac{1}{1000} & \frac{1}{1000} & & \frac{1}{1000} \end{pmatrix}$.

Thus: Y has the same distribution as X, but describes a totally different physical process.

(*) We can start seeing that different "processes" can be studied in a unified way, via probability models.

The binomial random variable

Context: If an experiment consists in n **independent** trials, each of which results in either a success, with probability p, or a failure, with probability $1-p$, then X, the random variable that counts the number of successes is said to be a binomial r.v. with parameters (n,p).

$$X \sim \begin{pmatrix} 0 & 1 & 2 & \dots & x & \dots & n \\ p_0 & p_1 & p_2 & \dots & p_x & \dots & p_n \end{pmatrix}, \text{ where}$$

$$p_x \overset{\text{def}}{=} P(X=x) = \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}$$

$$= f(x), \text{ the p.f. of } X.$$

- Note $f(x) = 0$, if $x \notin \{0, 1, \dots, n\}$.

• If X is as above we will write

$$X \sim \text{Bin}(n, p)$$

↑ "Has distribution / Is distributed"

Example • It is known that diskettes produced by a certain company will be defective with prob. 0.01, independtly of each other.

• The company sells the diskettes in packages of size 10 and offers a money back guarantee that at most 1 of the 10 diskettes in the package will be defective.

If someone buys 3 packages, what is the probability that he/she will return exactly one of them?

Solution

- Let Y be the random variable that counts the number of returned packages.

$Y \sim Bin(n=3, p)$.

$P(Y=1) = ?$

- Need $p = P(\text{A package is returned})$

$= P(\#\text{ of defective diskettes} \geq 2)$.

- Let X be the r.v. that counts the # of def. diskettes.

$X \sim Bin(n=10, p_1 = 0.01)$.

Therefore $p \equiv P(X \geq 2) = 1 - P(X \leq 1)$

$= 1 - [P(X=0) + P(X=1)]$

$= 1 - (0.99)^{10} - 10 \cdot (0.99)^9 \cdot (0.01)$.

- For this value of p, the final answer is:

$$P(Y=1) = \binom{3}{1} p(1-p)^2 = 3p(1-p)^2.$$

Continuous random variables

- $X \in A \subset \mathbb{R}$.

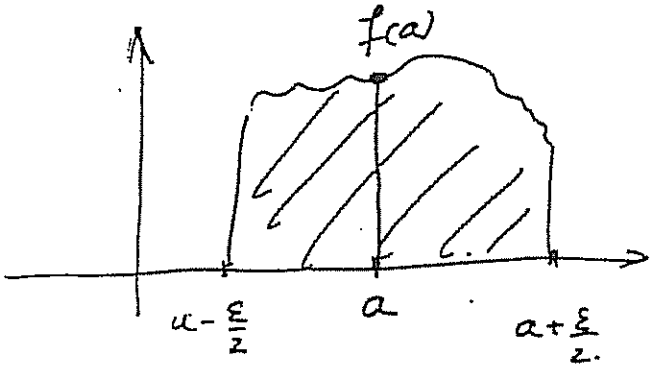
- X is a continuous r.v. if there exists $f(x) \geq 0$, for $x \in (-\infty, \infty)$ such that

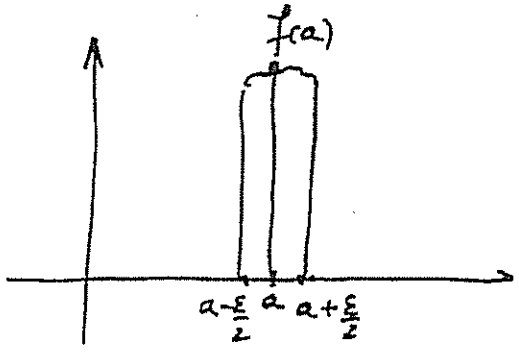
$$P(X \in B) = \int_B f(x)\,dx, \text{ for any } B \subset \mathbb{R}.$$

- $f(x)$ =: the density function of X =: p.d.f. of X

- $\int_{-\infty}^{\infty} f(x) = 1$.

- $P(a \leq X \leq b) = \int_a^b f(x)\,dx$.





- $P(a - \frac{\varepsilon}{2} \leq X \leq a + \frac{\varepsilon}{2}) = \int_{a-\frac{\varepsilon}{2}}^{a+\frac{\varepsilon}{2}} f(x)\,dx \simeq \varepsilon f(a)$

- Probability $(a \leq X \leq b)$ = the area underneath $f(x)$, for $a \leq x \leq b$.

- **Important** $P(X = a) = \int_{\{a\}} f(x)\,dx = 0,$

for any $a \in \mathbb{R}$.

Examples of continuous distributions

Def $X \sim \text{Unif}[a,b]$ if

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise.} \end{cases}$$

- Use X in problems in which the problem specifies only that "a point is chosen at random from $[a,b]$". If no further explanation/restrictions are imposed, the random mechanism can be modeled via a uniform random variable.

Application

Let $X \sim \text{Unif}[-1, 4]$. Then, the pdf is

$$f(x) = \begin{cases} \frac{1}{5}, & \text{for } -1 \le x \le 4 \\ 0, & \text{otherwise.} \end{cases}$$

$$P(0 \le X \le 2) = \int_0^2 f(x)\,dx = \frac{2}{5}.$$

Note: $P(0 < X < 2) = P(0 \le X < 2)$

$$= P(0 < X \le 2)$$

$$= P(0 \le X \le 2) = \frac{2}{5}.$$

This is not only valid for this example, and in general we make the following observation

Remark If $X \sim f(x)$, a continuous pdf, then:

$$P(c \le X \le d) = P(c < X < d)$$

$$= P(c \le X < d)$$

$$= P(c < X \le d), \text{ for}$$

any $c, d \in \mathbb{R}$.

Example The probability density function of X, the lifetime of a certain type of electronic device (measured in hours) is given by

$$f(x) = \begin{cases} \frac{C}{x^2}, & \text{if } x > 10 \\ 0, & \text{if } x \leq 10. \end{cases}$$

(a) Find C s.t. $f(x)$ can be used as a p.d.f.

(b) Find $P(X > 20)$.

Solution (a) $f(x)$ must satisfy: (i) $f(x) \geq 0$

(ii) $\int_{10}^{\infty} f(x)\,dx = 1.$

Therefore, $C \geq 0$ and we find it from:

$$C\int_{10}^{\infty} \frac{dx}{x^2} = 1 \iff \int_{10}^{\infty} \frac{dx}{x^2} = \frac{1}{C}.$$

$$\iff \left(-\frac{1}{x}\right)\Big|_{x=10}^{x=\infty} = 0 + \frac{1}{10} = \frac{1}{C} \implies$$

$$C = 10$$

(b) $P(X > 20) = \int_{20}^{\infty} \frac{10}{x^2} = 10\left(-\frac{1}{x}\right)\Big|_{20}^{\infty}$

$= \frac{10}{20} = \frac{1}{2}$.

(c) Assume further that we have 6 such devices. What is the probability that at least 3 of them function at least 20 hours?

• Let Y be the r.v. counting the # of devices that function ≥ 20 h.

Then $Y \sim Bin(n = 6, p)$, where

$p = P(X \geq 20) = P(X > 20) = \frac{1}{2}$.

Then, $P(Y \geq 3) = \sum_{i=3}^{6} \binom{6}{i} \left(\frac{1}{2}\right)^i \left(\frac{1}{2}\right)^{6-i}$

$= \left(\frac{1}{2}\right)^6 \sum_{i=3}^{6} \binom{6}{i}$.

The cumulative distribution function

Definition ... Let X be a r.v. The cumulative distribution function (c.d.f.) F of X is a function $F: \mathbb{R} \to [0,1]$, given by:

$$F(x) = P(X \leq x).$$

- All probability questions related to a r.v. X can be answered in terms of F. (see ① - ⑤ on next page).
- Although we distinguished (mainly for pedagogical reasons) between discrete and continuous random variables, the following properties hold for any F, irrespective of the type of the underlying X.

Properties of the c.d.f. F of a r.v. X

1. F is a nondecreasing function:
   If $a < b \Rightarrow F(a) \leq F(b)$.

2. $\lim_{b \to \infty} F(b) = 1$

3. $\lim_{b \to -\infty} F(b) = 0$

4. F is right continuous:
   a) For any b and any $b_n \downarrow b \Rightarrow$
$$F(b^+) =: \lim_{n \to \infty} F(b_n) = F(b)$$
   b) $F(a^+) =: \lim_{\substack{x \to a \\ x > a}} F(x) = F(a)$;

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1) $P(a < X \leq b) = F(b) - F(a)$

2) $P(a < X < b) = \lim_{n \to \infty} F(b - \frac{1}{n}) - F(a)$

3) $P(X < b) = \lim_{n \to \infty} F(b - \frac{1}{n}) =: F(b^-)$

4) $P(a \leq X < b) = \lim_{n \to \infty} F(b - \frac{1}{n}) - \lim_{n \to \infty} F(a - \frac{1}{n})$

5) $P(X = b) = P(X \leq b) - P(X < b) =$
$$= F(b) - \lim_{n \to \infty} F(b - \frac{1}{n}) = F(b) - F(b^-)$$

Definitions:

1) A sequence of events $\{E_n, n \geq 1\}$ is said to be increasing if

$$E_1 \subset E_2 \subset \ldots \subset E_n \subset E_{n+1} \subset \ldots$$

2) $\{E_n; n \geq 1\}$ is decreasing if:

$$E_1 \supset E_2 \supset \ldots \supset E_n \supset E_{n+1} \supset \ldots$$

3) If $\{E_n, n \geq 1\}$ is increasing:

$$\lim_{n \to \infty} E_n = \bigcup_{i=1}^{\infty} E_i$$

4) If $\{E_n; n \geq 1\}$ is decreasing:

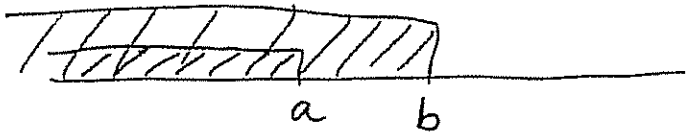
$$\lim_{n \to \infty} E_n = \bigcap_{i=1}^{\infty} E_i$$

Proposition If $\{E_n, n \geq 1\}$ is either an increasing or a decreasing sequence of events, then:

$$\lim_{n \to \infty} P(E_n) = P\left(\lim_{n \to \infty} E_n\right).$$

Note: This proposition says that probability is a continuous set function.

Proofs of 1 – 4



1. $F(a) = P(\{X \le a\})$
$F(b) = P(\{X \le b\})$

If $a < b \Rightarrow \{X \le a\} \subseteq \{X \le b\} \Rightarrow F(a) \le F(b)$.

2. $\lim_{b_n \nearrow \infty} F(b_n) = \lim_{b_n \nearrow \infty} P(X \le b_n)$

If $b_n \nearrow \Rightarrow \{(X \le b_1)\} \subset \{(X \le b_2)\} \subseteq \ldots \subset \{(X \le b_n)\} \subseteq \ldots$

$$\lim_{n\to\infty} \{(X \le b_n)\} = \bigcup_{n=0}^{\infty} \{(X \le b_n)\} = \{(X < \infty)\} =$$

$$= (-\infty < X < \infty) = S$$

Thus: $\lim_{b_n \nearrow \infty} F(b_n) = P\left(\lim_{n\to\infty} \{(X \le b_n)\}\right) = P(S) = 1$.

3. Exercise → Hint: use decreasing sequences, hence intersection and then the continuity of P.

Let $b_n \searrow b$. Since $b_n \searrow \Rightarrow b_1 > b_2 > b_3 > \ldots > b_n > \ldots$

$\Rightarrow \{X \le b_1\} \supset \{X \le b_2\} \supset \ldots \supset \{X \le b_n\} \supset \ldots \Rightarrow$

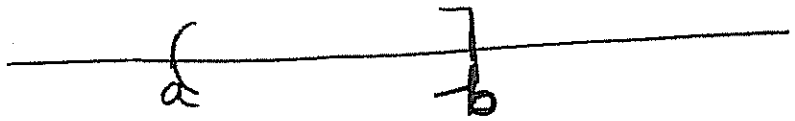
$$\Rightarrow \lim_{n\to\infty} \{X \le b_n\} = \bigcap_{n=1}^{\infty} \{X \le b_n\} = \{X \le b\}$$

Then: $\lim_{n\to\infty} P(\{X \le b_n\}) = P\left(\lim_{n\to\infty} \{X \le b_n\}\right) =$

$$= P(X < b) = F(b)$$

Proofs of ① – ⑤.

①

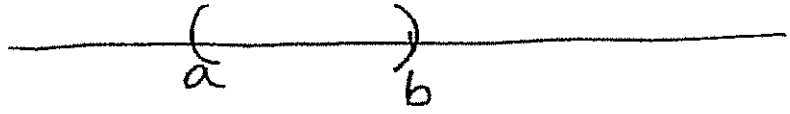


$(a, b] = (-\infty, b] - (-\infty, a] \Rightarrow$

$\Rightarrow P(a < X \leq b) = P(X \leq b) - P(X \leq a) =$

$= F(b) - F(a)$, for all $a < b$.

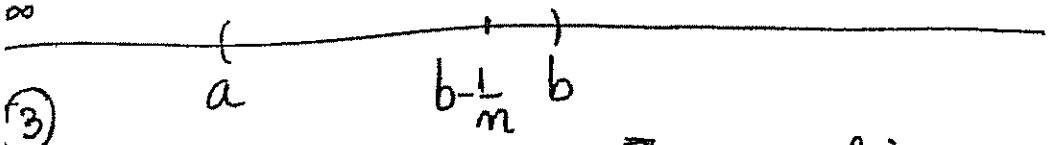
②



$(a, b) = (-\infty, b) - (-\infty, a] \Rightarrow$

$P(a < X < b) = P(X < b) - P(X \leq a) =$

$= P(X < b) - F(a)$

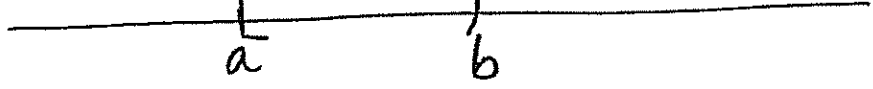


③

$(-\infty, b) = \bigcup_{n=0}^{\infty} (-\infty, b - \frac{1}{n}] \Rightarrow \lim_{n \to \infty} (-\infty, b - \frac{1}{n}]$

$P(X < b) = \lim_{n \to \infty} P(-\infty, b - \frac{1}{n}] = \lim_{n \to \infty} F(b - \frac{1}{n})$.

④



$[a, b) = (-\infty, b) - (-\infty, a) \Rightarrow$

$P(a \leq X < b) = P(X < b) - P(X < a)$.

Example The distribution function of the random variable X is given by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \le x < 1 \\ \frac{2}{3} & 1 \le x < 2 \\ \frac{11}{12} & 2 \le x < 3 \\ 1 & 3 \le x \end{cases}$$

A graph of $F(x)$ is presented in Figure 4.1. Compute (a) $P\{X < 3\}$, (b) $P\{X = 1\}$, (c) $P\{X > \frac{1}{2}\}$, and (d) $P\{2 < X \le 4\}$.

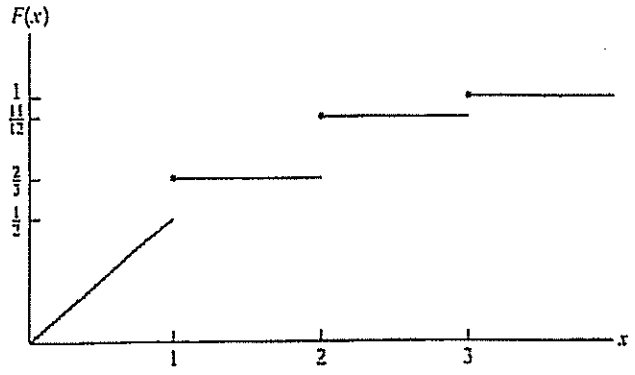


Figure 4.1 Graph of $F(x)$.

Solution

(a) $P\{X < 3\} = \lim_n P\left\{X \le 3 - \frac{1}{n}\right\} = \lim_n F\left(3 - \frac{1}{n}\right) = \frac{11}{12}$

(b) $P\{X = 1\} = P\{X \le 1\} - P\{X < 1\}$

$$= F(1) - \lim_n F\left(1 - \frac{1}{n}\right) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$$

(c). $P\left\{X > \frac{1}{2}\right\} = 1 - P\left\{X \le \frac{1}{2}\right\}$

$$= 1 - F\left(\frac{1}{2}\right) = \frac{3}{4}$$

(d) $P\{2 < X \le 4\} = F(4) - F(2)$

$$= \frac{1}{12}$$

The distribution function of a discrete r.v.

$$\times \begin{pmatrix} x_1 & x_2 & x_3 & \dots & x_n & \dots \\ p_1 & p_2 & p_3 & \dots & p_n & \dots \end{pmatrix} \qquad (x \downarrow)$$

$$\mathbb{P}(X \le x) = F(x) = \begin{cases} 0; & \text{if } x < x_1 \\ p_1; & \text{if } x_1 \le x < x_2 \\ p_1 + p_2; & \text{if } x_2 \le x < x_3 \\ \vdots & \\ p_1 + p_2 + \dots + p_{n-1}; & \text{if } x_{n-1} \le x < x_n \\ 1 \quad ; & \text{if } x \ge x_n. \end{cases}$$

Note Given $\{p_i\}_{i \ge 1}$ we can write out $F(x)$ as above. The reverse is also true: given $F(x)$, we can identify uniquely $p_1, \dots, p_n, \dots$.

Therefore: There is a one-to-one correspondence between $F(x)$, the c.d.f and $\{P(X = x)\}_{x \in \{x_1, \dots, x_n, \dots\}}$ the probability function.

Example Assume X has c.d.f. F given by:

$$F(x) = \begin{cases} 0; & x < 0 \\ \frac{1}{2}; & 0 \le x < 1 \\ \frac{3}{5}; & 1 \le x < 2 \\ \frac{4}{5}; & 2 \le x < 3 \\ \frac{9}{10}; & 3 \le x < 3.5 \\ 1; & x \ge 3.5 \end{cases}$$

It is clear then that X is discrete, with F connected to $P(X=x)$ as follows

$$F(x) = \begin{cases} 0; & x < 0 \\ p_1; & 0 \le x < 1 \\ p_1 + p_2; & 1 \le x < 2 \\ p_1 + p_2 + p_3; & 2 \le x < 3 \\ p_1 + p_2 + p_3 + p_4; & 3 \le x < 3.5 \\ p_1 + p_2 + p_3 + p_4 + p_5; & x \ge 3.5 \end{cases}$$

Therefore : $X \begin{pmatrix} 0 & 1 & 2 & 3 & 3.5 \\ p_1 & p_2 & p_3 & p_4 & p_5 \end{pmatrix}$,

where $p_1, \dots, p_5$ are the solutions of the system :

$$\begin{cases} p_1 = \frac{1}{2} \\ p_1 + p_2 = \frac{3}{5} \\ p_1 + p_2 + p_3 = \frac{4}{5} \\ p_1 + p_2 + p_3 + p_4 = \frac{9}{10} \\ p_1 + p_2 + p_3 + p_4 + p_5 = 1. \end{cases}$$

The cumulative distribution function
c.d.f of a continuous distribution

- Recall that for any r.v.

$$F(x) = \mathbb{P}(X \leq x).$$

Result If $X \sim f(x)$ is continuous, then

(1) $$F(x) = \int_{-\infty}^{x} f(t)\,dt,$$

(2) F is continuous at every x,

(3) $F'(x) = f(x)$, at each continuity point of f.

Example Let X be a r.v. with cdf given by

$$F(x) = \begin{cases} 0, & x < 0 \\ x^{\frac{2}{3}}, & 0 \le x \le 1 \\ 1, & \text{for } x > 1. \end{cases}$$

Find $f(x)$.

Solution: F is continuous over $\mathbb{R}$, and clearly differentiable on $(-\infty, 0)$, $(0,1)$ and $(1, \infty)$.

We now check if it's diff. at 0 and 1.

$$\lim_{\substack{x \to 0 \\ x > 0}} \frac{F(x) - F(0)}{x - 0} = \lim_{\substack{x \to 0 \\ x > 0}} \frac{x^{\frac{2}{3}} - 0}{x}$$

$$= \lim_{\substack{x \to 0 \\ x > 0}} \frac{1}{x^{\frac{1}{3}}} = \infty \quad \Rightarrow$$

Not diff. at zero.

check at 1:

$$\lim_{\substack{x \to 1 \\ x < 1}} \frac{F(x) - 1}{x - 1} = \lim_{\substack{x \to 1 \\ x < 1}} \frac{x^{\frac{2}{3}} - 1}{(x^{\frac{1}{3}} - 1)(x^{\frac{2}{3}} + x^{\frac{1}{3}} + 1)}$$

$$= \lim_{\substack{x \to 1 \\ x < 1}} \frac{x^{\frac{1}{3}} + 1}{x^{\frac{2}{3}} + x^{\frac{1}{3}} + 1} = \frac{2}{3}$$

$$\lim_{\substack{x\to 1\\ x>1}} \frac{F(x)-1}{x-1} = \lim_{\substack{x\to 1\\ x>1}} \frac{0}{x-1} = 0 \neq \frac{2}{3},$$

not diff at 1.

Thus
$$f(x) = \begin{cases} 0, & x \leq 0 \\ \frac{2}{3} x^{-\frac{1}{3}}, & 0 < x < 1 \\ 0, & \text{for } x \geq 1. \end{cases}$$

Note The values of f at $x=0$ and $x=1$ can be assigned arbitrarily, since changing the expression of $f(x)$ at a finite number of points will not change the values of the probabilities that can be calculated using f: $\mathbb{P}(X \leq 0) = \mathbb{P}(X < 0)$

$$= \int_{-\infty}^{0} f(x)\,dx, \text{ for instance.}$$

For uniformity, we defined $f(0) = 0$
$f(1) = 0$.

The quantile function

Example (Motivation of notation and definition)

- Let X = the amount of snow that will fall tomorrow.
- Place a bet: $X \le x_0$, you win
  $X > x_0$, you lose.
- To make the bet fair you need to decide on a value x_0 s.t.

$$P(X \le x_0) = P(X > x_0) = \frac{1}{2}.$$

- Question: How to find such an x_0?
- Answer: Assume X has c.d.f F, and you know F.

- If you know F, then x_0 is found by solving $F(x) = \frac{1}{2}$ for x.
- If F had an inverse, say F^{-1}, then $x_0 = F^{-1}\left(\frac{1}{2}\right)$.
- If you didn't want to make the bet fair, you could have, instead, searched for, say, x_1 s.t.

$$\mathbb{P}(X \leq x_1) = \frac{2}{3} \quad \text{and, as above,}$$

$$x_1 = F^{-1}\left(\frac{2}{3}\right).$$

- We would call x_0 the 50% percentile of X, as 50% of the distr. of X (as measured by the c.d.f F) is at or below x_0.

- Similarly, we would call x_1 the 66% percentile of X.
- However, the inverse of F doesn't always exist, but we can still define percentiles as below.

Definition - Let X be a r.v. with c.d.f. F.

- Let $p \in (0,1)$.
- The smallest value of x s.t. $F(x) \geq p$ is denoted $F^{-1}(p)$ and called the p quantile of X or the $100p$% percentile of X.
- The function F^{-1} defined on $(0,1)$ is called the quantile function.
- F^{-1} is abuse of notation. F^{-1} as above is NOT always the inverse of F.

Example 3.3.7

Value at Risk. The manager of an investment portfolio is interested in how much money the portfolio might lose over a fixed time horizon. Let X be the change in value of the given portfolio over a period of one month. Suppose that X has the p.d.f. in Fig. 3.7. The manager computes a quantity known in the world of risk management as *Value at Risk* (denoted by VaR). To be specific, let $Y = -X$ stand for the loss incurred by the portfolio over the one month. The manager wants to have a level of confidence about how large Y might be. In this example, the manager specifies a probability level, such as 0.99 and then finds y_0, the 0.99 quantile of Y. The manager is now 99% sure that $Y \leq y_0$, and y_0 is called the VaR. If X has a continuous distribution, then it is easy to see that y_0 is closely related to the 0.01 quantile of the distribution of X. The 0.01 quantile x_0 has the property that $\Pr(X < x_0) = 0.01$. But $\Pr(X < x_0) = \Pr(Y > -x_0) = 1 - \Pr(Y \leq -x_0)$. Hence, $-x_0$ is a 0.99 quantile of Y. For the p.d.f. in Fig. 3.7, we see that $x_0 = -4.14$, as the shaded region indicates. Then $y_0 = 4.14$ is VaR for one month at probability level 0.99. ◀

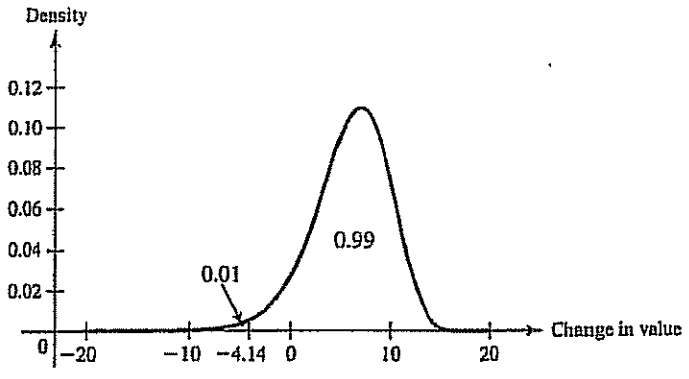


Figure 3.7 The p.d.f. of the change in value of a portfolio with lower 1% indicated.

Result If the cdf F of X is continuous and one-to-one, the inverse F^{-1} of F exists and equals the quantile function of X.

- Example 3.3.~~7~~.

Terminology: 1/2 quantile = median
1/4 quantile = lower quartile
3/4 quartile = upper quartile.

Reading exercise: Quantiles of discrete distributions.